

Jitin Kumar

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RESEARCH INTERESTS

Applied machine learning, NLP and AI for climate action and education: predictive modelling, explainable and personalised AI systems.

EDUCATION

B. Tech, Computer Science Engineering (AI) - CSJM University (UIET), Kanpur Aug 2024 - Jun 2028

- CGPA 8.71 / 10; consistently in the top 15% of the B. Tech CSE (AI)
- Relevant coursework: Machine Learning, Computer Vision, Data Structures, Object-Oriented Programming, Operating Systems

RESEARCH AND APPLIED EXPERIENCE

Green Skills and Applied AI for Climate Action Intern - 1M1B Jun 2026 - Aug 2026

- Completed a 70+ hour programme on applying AI to climate action and green skills, with Microsoft Elevate, MeitY Startup Hub and UP Pragya
- Built and deployed SkillGreen, a full-stack ESG career-readiness predictor (FastAPI, scikit-learn, React); currently extending it with resume parsing, model explainability and skill-gap analysis

Data Science and Analytics Intern - Tamizhan Skills (Remote) Jun 2025 - Jul 2025

- Ran end-to-end data pipelines (ingestion, cleaning, EDA) on 5+ datasets using Pandas, Matplotlib and Seaborn
- Reduced dataset inconsistencies by ~25% through systematic outlier detection and imputation, improving downstream model reliability
- Cut data-manipulation time by ~40% by replacing iterative loops with vectorised NumPy operations
- Built 3 interactive Plotly dashboards that reduced ad-hoc stakeholder report requests

Machine Learning Intern - TechKriti, IIT Kanpur Jan 2025 - Feb 2025

- Implemented 2 supervised models (NLP classification and regression) covering training, evaluation and iteration
- Reduced training time by ~20% through feature selection and Scikit-learn preprocessing optimisations
- Ranked among the top 3 intern submissions at the internal showcase; presented results to an IIT Kanpur faculty panel

AWS APAC Solutions Architecture - Forage Virtual Program Jan 2025

- Designed a scalable AWS Elastic Beanstalk hosting architecture for a simulated client brief and wrote a technical recommendation for a non-technical audience

PROJECTS

SkillGreen: ESG Career-Readiness Predictor [GitHub](#) | [Live](#) 2026

- Predicts ESG (Environmental, Social, Governance) career readiness from a professional's profile using a scikit-learn model, a FastAPI + Pydantic backend and a React frontend; built during the 1M1B internship
- Deployed with the frontend on Vercel and the backend and database on Render, Docker-based setup; adding resume parsing, explainability and a skill-gap engine
- Stack: Python, FastAPI, Pydantic, Scikit-learn, React, Vite, PostgreSQL, Docker

RiskLens: Insurance Premium Risk Predictor [GitHub](#) 2026

- Predicts a Low / Medium / High insurance premium risk category from a user's profile, with feature engineering built into the model's input pipeline and served through FastAPI
- Stack: Python, FastAPI, Feature Engineering

ModelServe: FastAPI Serving Layer for ML Models [GitHub](#) 2026

- Built a FastAPI service that deploys machine learning models as REST APIs, with preprocessing, prediction and evaluation endpoints and automatic API docs
- Stack: Python, FastAPI, REST APIs, Jupyter

Student Performance Analytics Dashboard [GitHub](#) 2025

- Merged 3 data sources (grades, attendance, login activity) into one pipeline feeding a faculty-facing at-risk dashboard
- Trained a Random Forest classifier with 87% accuracy to identify low-performing students before end of term; per-student improvement suggestions were adopted by faculty for early intervention
- Stack: Python, Pandas, Scikit-learn, Matplotlib, Jupyter

TECHNICAL SKILLS

Languages: Python, C/C++

ML / AI: Linear Regression, Random Forest, Feature Engineering, NLP, Predictive Modelling, Computer Vision (coursework)

Libraries: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, Plotly

Tools: Jupyter, Git, VS Code, Docker, FastAPI, Pydantic, React, AWS Elastic Beanstalk

CERTIFICATIONS AND ACTIVITIES

- Machine Learning Certification, TechKriti, IIT Kanpur (Feb 2025)
- Data Science and Analytics Certification, Tamizhan Skills (Jul 2025)
- Machine Learning with Python, FreeCodeCamp (Dec 2025)
- Green Skills and Applied AI for Climate Action Internship, 1M1B (Aug 2026)
- National Service Scheme (NSS): active volunteer